RESEARCH / PROJECT ETHICS REVIEW FORM For Thesis / Capstone / Dissertation Proposals (*ver.Oct2024c*)
Names of Student Researcher(s): **Adrian Rafael Bernandino** **Anton Miguel Borromeo** **Russell Emmanuel Galan** **Reinman Geoffrey Ganayo**
Research/Thesis/Capstone Title: Modernizing 3D Graphics Rendering for Interactive Taal Volcano Plume Simulation
College: College of Computer Studies
Department: Software Technology
Course: BS Computer Science major in Software Technology
Expected Duration of the Project: from: Aug 6, 2026 to Aug 6, 2027

PROTOCOL:

(*To be filled up by proponents; Use the Ethics Checklists as guides in determining areas for ethical concern/consideration. Attach and properly reference the accomplished ethics checklists and supporting documents.)*

		Reference Document (include page number)
Target participants	10 to 20 voluntary participants consisting of collegiate computer science students, graphics programmers, or software developers.	Scope and Limitations (page x-x)
Data to be used / collected	**Technical Data:** Empirical hardware performance logs including Frame Rate (FPS), CPU/GPU resource utilization percentage, and frame rendering times (ms). **Human Data:** Quantitative usability scores via the System Usability Scale (SUS) questionnaire and qualitative feedback regarding visual artifacts or rendering anomalies. No personally identifiable information (PII) such as names, student numbers, or email addresses will be tied to the responses.	Scope and Limitations (page x-x)
Brief procedure	1. **Multi-API Execution:** Run the legacy OpenGL *AnitoPlume* alongside the newly developed modern low-level explicit API implementations (such as Vulkan or DirectX 12) under identical hardware	Scope and Limitations (page x-x)

	environments to capture raw performance data. 2. **Informed Consent:** Present participants with an Informed Consent Form (ICF) outlining the evaluation's scope, hardware interactions, and data privacy measures before any testing occurs. 3. **Cross-API Usability Testing:** Participants will interact with the different API implementations of the interactive 3D plume simulator to evaluate real-time responsiveness and stability. 4. **Questionnaire Evaluation:** Participants will complete an anonymous post-test SUS questionnaire to measure and compare usability and visual consistency across the target graphics backends.	
Potential risks		
Applicable Terms of Use or Licensing Agreement of Data, Models and/or Tools	**AnitoPlume Source Code:** Used with explicit permission and under the supervision of the thesis adviser, Neil Del Gallego. **Libraries:** Eigen (Mozilla Public License v2.0) and dearIMGUI (MIT License). **Graphics Toolkits & SDKs:** Khronos Vulkan SDK, Microsoft DirectX 12 SDK, and Khronos OpenGL specifications.	Scope and Limitations (page x-x)
Steps to safeguard the participants and/or data	**Participant Safety:** Testing sessions will be kept brief (under 20 minutes) to prevent eye strain, and participants may withdraw from the usability test at any point without penalty. **Data Safety:** All performance logs and usability questionnaire responses will be fully anonymized using alphanumeric identifiers (e.g., User01, User02). Survey sheets and digital spreadsheets will be secured in a password-protected cloud storage repository accessible only to the primary student researchers, to be permanently deleted upon completion of the thesis defense	Scope and Limitations (page x-x)
Other concerns and considerations	None at this point in time	

PANEL'S ASSESSMENT/COMMENTS:

PANEL'S DECISION:

- € **Approved** (*Proponent/s may proceed with the study/data collection*)
- € **Modifications Required** (*Proponent/s should address ethical issues, follow the panel's recommendations for required modifications, and resubmit Ethics Review Form for approval before proceeding with the study or before starting with data collection*)
- € **For Full Review** (*Proponent/s should submit the proposal to the University Research Ethics Review Committee (RERC) and obtain a research ethics clearance before proceeding with the study or before starting with data collection*)

JUSTIFICATION:
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Name and Signature of Adviser/Mentor

Date: ______________________

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Name and Signature of Panel Member **Name and Signature of Panel Member**

Date: ______________________ **Date:** ______________________

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Name and Signature of Panel Member

Date: ______________________